Aarya Patil

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Education

- **B. Tech in Computer Engineering, K. J. Somaiya College of Engineering** (2023 – 2027)
  Current GPA: 9.32

Experience

- **AI/ML Intern, Amnex Infotechnologies** (June 2026 – August 2026)
  - Architected **AIRA**, a self-hosted enterprise RAG platform over **AlloyDB + pgvector** (26-table schema) with **BGE-M3** embeddings and **Gemma 3 27B on vLLM**, featuring hybrid HNSW vector + GIN full-text search over 21 ingested HR policy documents with incremental SHA-256 change detection.
  - Built a secured **13-tool agent orchestrator** with constrained JSON tool-calling (Gemma 3 27B, max 4 round-trips), server-side RBAC identity injection, confidence-gated HR escalation, and a **hash-chained audit log**.
- **Computer Vision Intern, Fostride** (March 2026 – April 2026)
  - Built an **automated waste classification pipeline** for a robotic sorter system, engineering a **YOLO + DINOv2** pipeline to resolve domain generalisation across capture environments.
  - Developed and iterated an **autolabelling pipeline** end-to-end across segmentation, classification, and inference stages, targeting production deployment on the **Jetson Nano**.
- **AI/ML Intern, PricewaterhouseCoopers (PwC) India** (May 2025 – August 2025)
  - Built CV pipeline **(YOLOv8, OpenCV, Pytesseract)** digitizing legacy land records at ≈ 1 page/sec with **85-90% redaction accuracy** on 20+ year old degraded scans.
  - Engineered NLP system automating monthly report analysis from **60+ State Govt. departments**, reducing manual effort from 2 days to 30 min.
- **Data Acquisition Engineer, Orion Racing, KJSCE** (Feb 2024 – Oct 2024)
  - Architected telemetry pipelines streaming **25+ CAN sensors** to dashboards with **sub 200ms latency** for real-time powertrain & suspension monitoring.
  - Built anomaly detection across **100k+ datapoints/lap** leading to **5× faster** post-race diagnostics.

Projects

- **TrafficOS – Optimizing Traffic**
  - Built traffic congestion forecasting system using **Temporal Fusion Transformer (TFT)** with grid-based heatmap visualization.
  - Implemented **game-theoretic route optimization** to distribute traffic and reduce congestion across road networks, solving the **algorithmic gridlock problem** at source.
- **Indexial – Hybrid SQL + RAG Query Engine**
  - Built document QA system routing queries to **SQL/RAG/hybrid paths** via intent classification with cross-document semantic search using **Jina AI embeddings + pgvector**.
  - Designed session-based memory with automatic cleanup and **persistent table registry** maintaining database schemas.
- **Estro – WebMCP Photo Editor**
  - Built a local-first photo editor in **React, TypeScript, and IndexedDB** with non-destructive editing, layers, masks, and revision history.
  - Exposed **56 WebMCP tools** for validated, conflict-checked edits with per-transaction undo and agent-visible UI feedback.

Additional Credentials

- **Competitive Programming:** Rated 1600+ LeetCode, ranked **Pupil** on CodeForces.
- **McKinsey.org Forward Program**: Structured problem-solving, communication, digital foundations (Dec 2025).
- **DeepLearning.AI Specialization (4 courses)**: Neural Networks and Deep Learning, Improving Deep Neural Networks, Structuring Machine Learning Projects, Convolutional Neural Networks (CNNs)

Knowledge and Skills

- **ML/AI:** RAG, Transformers, CV, NLP, TFT
- **Languages:** Python, C++, Java, SQL, JavaScript
- **Frameworks/Tools:** PyTorch, TensorFlow, Hugging Face, FastAPI, Flask, Next.js, React, Git, Docker
- **Databases/Infra:** Supabase (pgvector), Vector DBs, PostgreSQL, Redis